Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Lucas Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
11204 Fulkerth RD	Turlock	Stanislaus	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1938

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X022-X037-X010-XXXX

B. OPERATORS

Lucas, Michael Edward

Operator name: Lucas, Michael Edward

Telephone no.: (209) 667-5749 Landline, (209) 531-6792 Cellular

Mailing Address Number and Street	City	State	Zip Code
1030 Bystrum RD	Turlock	CA	95380

This operator is responsible for paying permit fees.

Lucas, Thomas Michael

Operator name: Lucas, Thomas Michael

Telephone no.: (209) 669-9059 Landline, (209) 380-8319 Cellular

Mailing Address Number and Street	City	State	Zip Code
854 Bystrum RD	Turlock	CA	95380

C. OWNERS

Lucas, Michael Edward

Legal owner name:	Lucas, Michael Edward	Telephone no.:	(209) 667-5749	(209) 531-6792
			Landline	Cellular
1030 Bystrum RD	Turlock		CA	95380
Mailing Address Number and Street	City		State	Zip Code

This owner is responsible for paying permit fees.

Lucas, Thomas Michael

Legal owner name:	Lucas, Thomas Michael	Telephone no.:	(209) 669-9059	(209) 380-8319
			Landline	Cellular
854 Bystrum RD	Turlock		CA	95380
Mailing Address Number and Street	City		State	Zip Code

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	683	0	0	0	0	0
Maximum number	683	0	0	0	0	0
Average number	171	0	0	0	0	0
Avg live weight (lbs)	1,400	0	0	0		

Predominant milk cow breed: Holstein

Average milk production: 80 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 4,588.32 *tons per reporting period*

Total nitrogen from manure: 59,973.92 *lbs per reporting period*

After ammonia losses (30% loss applied): 41,981.74 *lbs per reporting period*

Total phosphorus from manure: 10,191.48 *lbs per reporting period*

Total potassium from manure: 33,871.55 *lbs per reporting period*

Total salt from manure: 80,515.35 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 7,185,750 *gallons*

Total nitrogen generated: 17,725.68 *lbs*

Total phosphorus generated: 4,593.33 *lbs*

Total potassium generated: 24,825.54 *lbs*

Total salt generated: 140,917.95 *lbs*

	2,125,500	*gallons applied*
+	5,060,250	*gallons exported*
-	0	*gallons imported*
=	7,185,750	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - Blaker	Ground water
Domestic Well - Dilworth	Ground water
Domestic Well - Mike	Ground water
Domestic Well - Milk Barn	Ground water
Domestic Well - Norman	Ground water

Source Description	Type
Domestic Well - Old Milk Barn	Ground water
Domestic Well - Pit	Ground water
Domestic Well - Tom	Ground water
Irrigation Water - Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/15/2023	Corral solids	2,700.00 *ton*	Dry-weight	6.3		10,600.00	6,200.00	16,700.00		0.00

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
08/12/2023	Process wastewater	2,671,500.00 *gal*	294.00	145.00	0.00	1.60	76.60	414.00		2,350
08/15/2023	Process wastewater	1,560,000.00 *gal*	294.00	145.00	0.00	1.60	76.60	414.00		2,350
08/17/2023	Process wastewater	828,750.00 *gal*	294.00	145.00	0.00	1.60	76.60	414.00		2,350

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	53,633.88	31,370.76	84,498.66	0.00
Process wastewater	12,482.53	3,234.65	17,482.30	99,235.30
Total exports for all materials	66,116.41	34,605.41	101,980.96	99,235.30

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
15 Acre	15	15	0	none	X022-X037-X010-XXXX
30 Acre	29	29	0	none	X022-X040-X002-XXXX
Blaker	39	39	0	none	X022-X040-X012-XXXX
Dillworth	39	39	0	none	X022-X037-X007-XXXX
McPhearson	39	39	0	none	X022-X040-X011-XXXX
Mike's	34	34	0	none	X022-X037-X008-XXXX
Morgan	39	39	0	none	X022-X040-X013-XXXX
Pit	34	34	2	both	X022-X037-X010-XXXX
Totals for areas that were used for application	34	34	2		
Totals for areas that were not used for application	234	234	0		
Land application area totals	268	268	2		

B. CROPS AND HARVESTS

Pit

Field name: Pit

11/21/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 34 Plant date: 11/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/18/2023	522.00 *ton*	Dry-weight		72.6	24,000.00	5,100.00	57,600.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.35	201.92	42.91	484.61	841.34

Pit

05/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 34 Plant date: 05/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/30/2023	1,120.00 *ton*	Dry-weight		67.9	11,800.00	2,350.00	18,800.00		11.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.94	249.55	49.70	397.59	2,347.45

NUTRIENT BUDGET

A. LAND APPLICATIONS

Pit - 11/21/2022: Oats, silage-soft dough

Field name: Pit

Crop: Oats, silage-soft dough Plant date: 11/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	33.25	8.62	46.56	264.31	458,250.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	38.18	5,017,575.00 *gal*
Application event totals		33.25	8.62	46.56	302.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	31.83	8.25	44.58	253.07	438,750.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.92	4,983,900.00 *gal*
Application event totals		31.83	8.25	44.58	290.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	146.06	85.43	230.12	0.00	250.00 *ton*
Application event totals		146.06	85.43	230.12	0.00	

Pit - 05/02/2023: Corn, silage

Field name: Pit

Crop: Corn, silage Plant date: 05/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Pit - 05/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	217.05	77.22	244.19	0.00	400.00 *ton*
Application event totals		217.05	77.22	244.19	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.56	5,724,750.00 *gal*
Application event totals		0.00	0.00	0.00	43.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.66	4,950,225.00 *gal*
Application event totals		0.00	0.00	0.00	37.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.00	7.52	40.62	230.57	399,750.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.10	4,613,475.00 *gal*
Application event totals		29.00	7.52	40.62	265.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.85	4,579,800.00 *gal*
Application event totals		0.00	0.00	0.00	34.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.95	7.76	41.94	238.07	412,750.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.92	4,983,900.00 *gal*
Application event totals		29.95	7.76	41.94	275.99	

Pit - 05/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	30.18	7.82	42.27	239.94	416,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.10	4,613,475.00 *gal*
Application event totals		30.18	7.82	42.27	275.05	

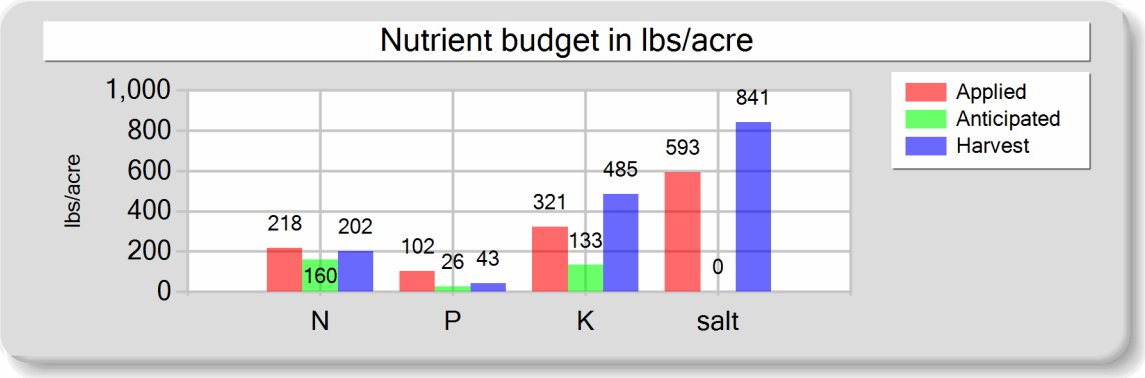
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.38	4,781,850.00 *gal*
Application event totals		0.00	0.00	0.00	36.38	

B. NUTRIENT BUDGET

Pit - 11/21/2022: Oats, silage-soft dough

Field name: Pit | Crop: Oats, silage-soft dough | Plant date: 11/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	146.06	85.43	230.12	0.00
Process wastewater	65.08	16.86	91.15	517.38
Fresh water	0.00	0.00	0.00	76.10
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	218.14	102.30	321.26	593.48
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	201.92	42.91	484.61	841.34
Nutrient balance	16.22	59.39	-163.35	-247.87
Applied to removed ratio	1.08	2.38	0.66	0.71

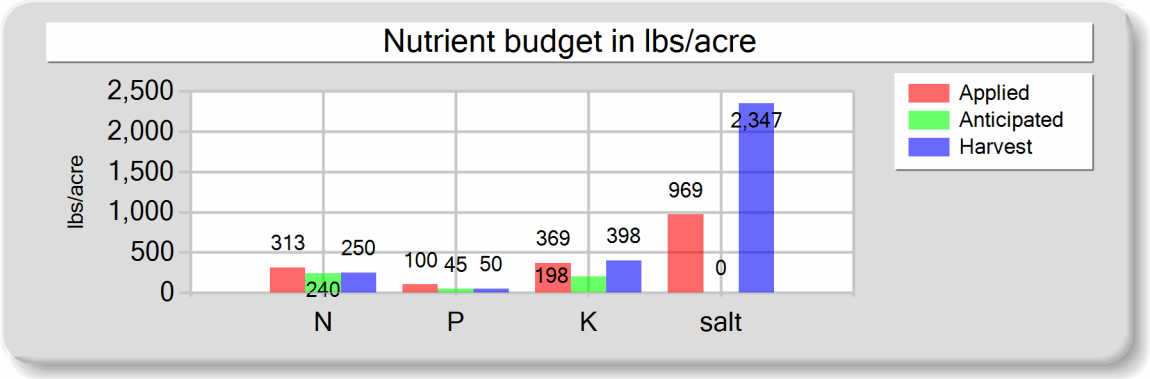
Fresh water applied	
10,001,475.00	*gallons*
368.32	*acre-inches*
10.83	*inches/acre*

Process wastewater applied	
897,000.00	*gallons*
33.03	*acre-inches*
0.97	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pit - 05/02/2023: Corn, silage

Field name: Pit | Crop: Corn, silage | Plant date: 05/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	217.05	77.22	244.19	0.00
Process wastewater	89.13	23.10	124.83	708.58
Fresh water	0.00	0.00	0.00	260.58
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	313.18	100.32	369.02	969.16
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	249.55	49.70	397.59	2,347.45
Nutrient balance	63.64	50.62	-28.57	-1,378.29
Applied to removed ratio	1.26	2.02	0.93	0.41

Fresh water applied	
34,247,475.00	*gallons*
1,261.22	*acre-inches*
37.09	*inches/acre*

Process wastewater applied	
1,228,500.00	*gallons*
45.24	*acre-inches*
1.33	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure
Sample date: 05/24/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 11.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,400.00	3,700.00	11,700.00							
DL	100.00	100.00	100.00							

Solid Manure

Sample and source description: Solid Manure
Sample date: 10/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 6.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,600.00	6,200.00	16,700.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water
Sample date: 02/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	294.00	145.00	0.00	1.60	76.60	414.00								4,740.00	2,350
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon water

Sample and source description: Lagoon water

Sample date: 06/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	167.00	120.00	0.00	0.40	85.70	805.00								4,510.00	2,000
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	55.30	43.70	0.00	0.40	16.10	93.60								1,350.00	750
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	69.30	55.60	0.00	0.40	16.60	93.00								1,290.00	675
DL	1.00	0.50	0.01	0.10	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - Blaker

Domestic Well - Blaker

Blaker

Sample description: Blaker

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			43.90								1,160.00	
DL			0.10								10.00	

Domestic Well - Dilworth

Dilworth

Sample description: Dilworth

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			29.50								1,330.00	
DL			0.10								10.00	

Domestic Well - Mike

Mike

Sample description: Mike

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			29.70								1,350.00	
DL			0.10								10.00	

Domestic Well - Milk Barn

Domestic Well - Milk Barn

Milk Barn

Sample description: Milk Barn

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			41.90								1,330.00	
DL			0.10								10.00	

Domestic Well - Norman

Norman

Sample description: Norman

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			24.80								834.00	
DL			0.10								10.00	

Domestic Well - Old Milk Barn

Old Milk Barn

Sample description: Old Milk Barn

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			42.00								1,350.00	
DL			0.10								10.00	

Domestic Well - Pit

Domestic Well - Pit

Pit

Sample description: Pit

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			89.90								2,020.00	
DL			0.10								10.00	

Domestic Well - Tom

Tom

Sample description: Tom

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			18.00								1,160.00	
DL			0.10								10.00	

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 08/09/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00										41.30	31
DL	0.10										10.00	10

D. SOIL ANALYSES

Pit

Pit Field

Sample and source description: Pit Field

Sample date: 11/07/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			74.10				
DL			2.00				

E. PLANT TISSUE ANALYSES

Pit - 11/21/2022: Oats, silage-soft dough

Pit

Sample and source description: Pit

Sample date: 04/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,000.00	5,100.00	57,600.00		
DL	100.00	100.00	100.00		

Pit - 05/02/2023: Corn, silage

Pit

Sample and source description: Pit

Sample date: 08/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	2,350.00	18,800.00		
DL	100.00	100.00	100.00		

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

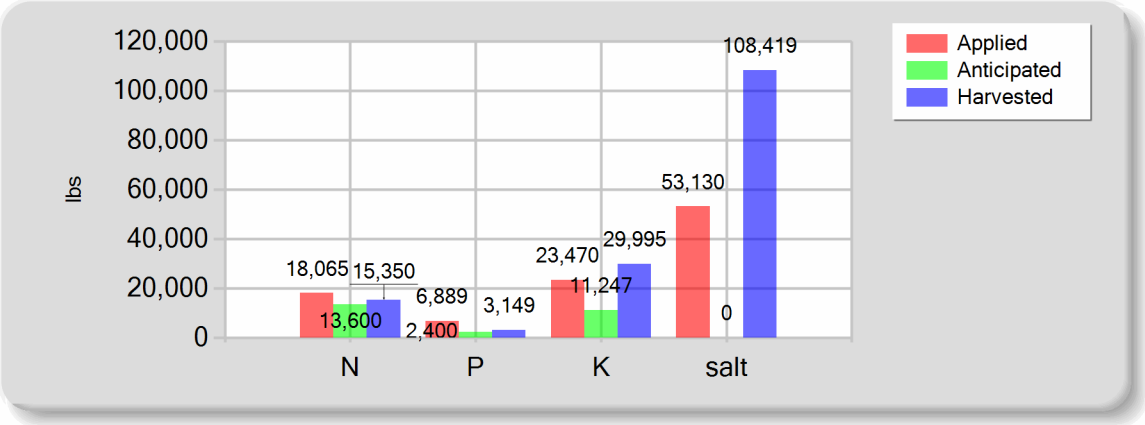
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

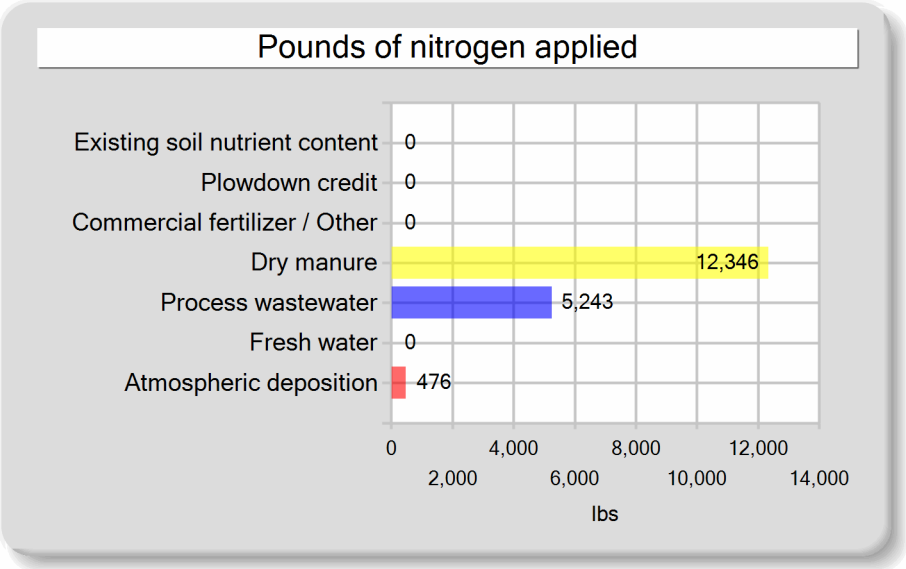
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

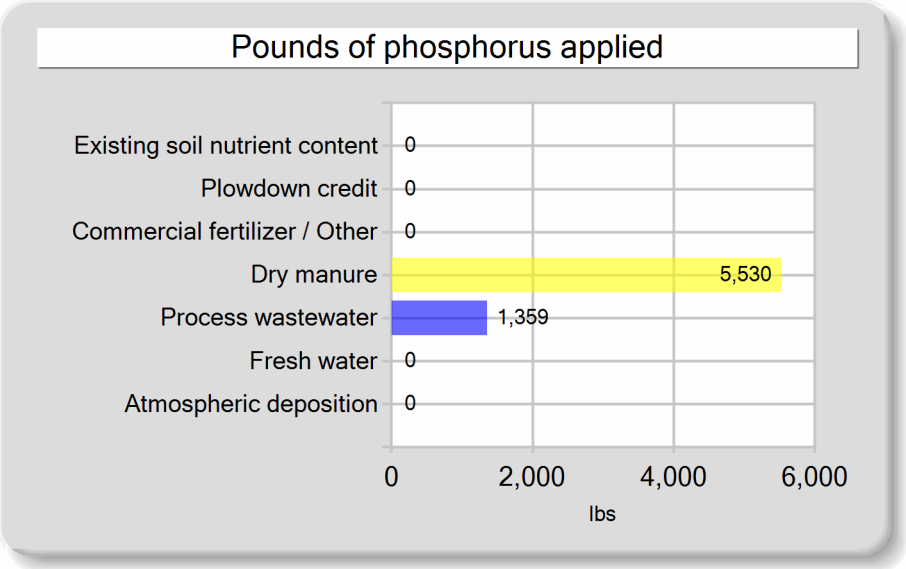
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	12,345.94	5,530.22	16,126.27	0.00
Process wastewater	5,243.15	1,358.68	7,343.24	41,682.65
Fresh water	0.00	0.00	0.00	11,446.98
Atmospheric deposition	476.00	0.00	0.00	0.00
Total nutrients applied	18,065.09	6,888.90	23,469.51	53,129.63
Anticipated crop nutrient removal	13,600.00	2,400.40	11,247.20	0.00
Actual crop nutrient removal	15,350.02	3,148.63	29,994.78	108,419.04
Nutrient balance	2,715.07	3,740.27	-6,525.27	-55,289.41
Applied to removed ratio	1.18	2.19	0.78	0.49

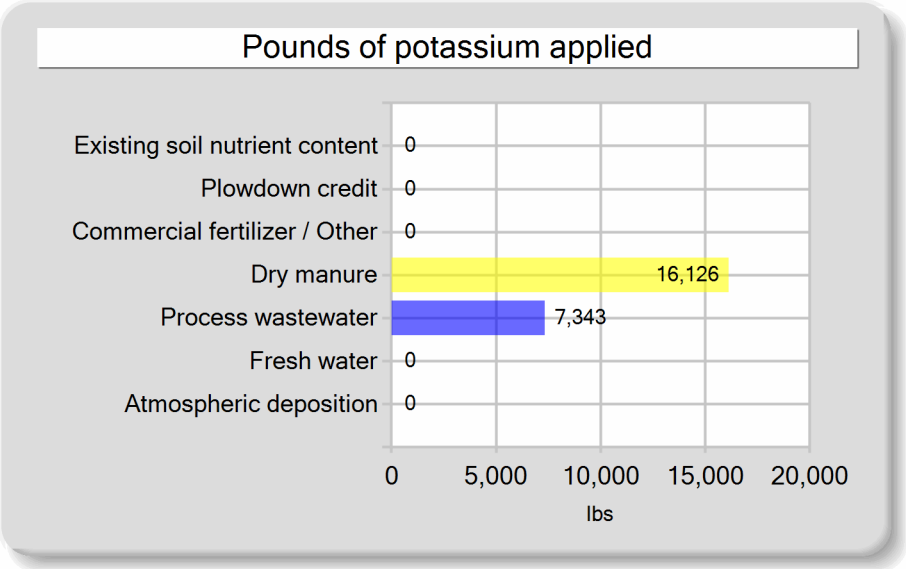
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

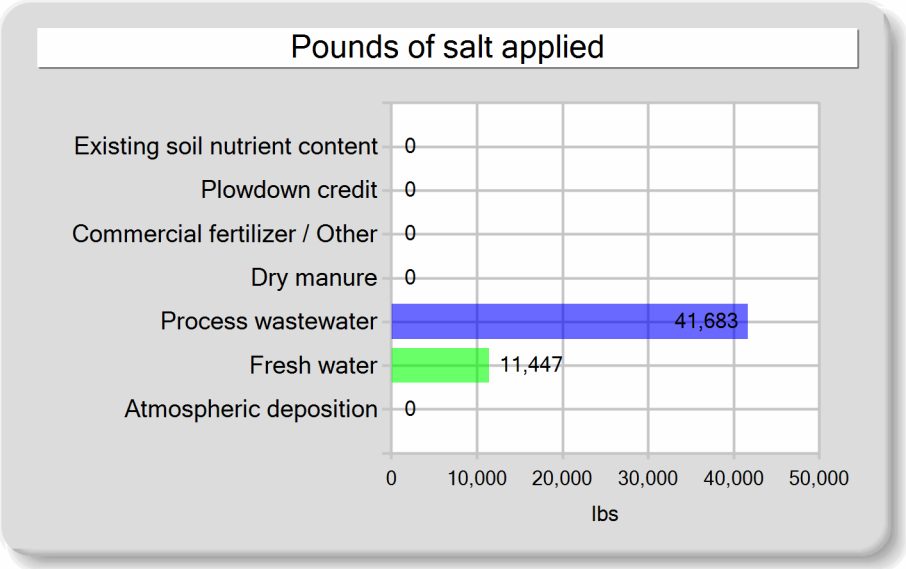


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

***Soluble phosphorus reported as PO4-P (OLSEN)

Cattle were sold on 3/30/2023 and the facility remained EMPTY for the remainder of 2023*

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Tom Lucas
Tom Lucas (Mar 28, 2024 15:10 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Thomas Michael Lucas	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
28/03/24	
DATE	DATE

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: **Mike or Tom Lucas**

Name of Dairy Facility: **Lucas Dairy**

Facility Address: **11204 Fulkerth Road, Turlock, CA 95380**

Contact Person Name and Phone Number: **Mike or Tom Lucas (209) 277-6499**

Manure Hauler Information:

Name of Hauling Company/Person: **Lucas Dairy**

Address of Hauling Company /Person: **11204 Fulkerth Road, Turlock, CA 95380**

Contact Person and Phone Number: **Mike or Tom Lucas (209) 277-6499**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Farmer** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
John Bailey	**1312 N Morgan Road**	**Turlock**	**95380**	**(209) 678-3427**

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
Stanislaus County			**APN 0022-0040-0001-0000**

Dates Hauled: **3/5/23, 3/14/23, 4/9/23, 7/22/23, 8/4/23 & 8/17/23**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____Tons

Manure Solids Content (if amount reported in tons): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **828,750** Gallons

Method used to determine volume of process wastewater: _____

Total Run Time x Pump Output GPM

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mike Lucas (Mar 28, 2024 15:12 PDT) Date: 28/03/24

Hauler's Signature: Mike Lucas (Mar 28, 2024 15:12 PDT) Date: 28/03/24

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: **Mike or Tom Lucas**

Name of Dairy Facility: **Lucas Dairy**

Facility Address: **11204 Fulkerth Road, Turlock, CA 95380**

Contact Person Name and Phone Number: **Mike or Tom Lucas (209) 277-6499**

Manure Hauler Information:

Name of Hauling Company/Person: **Lucas Dairy**

Address of Hauling Company /Person: **11204 Fulkerth Road, Turlock, CA 95380**

Contact Person and Phone Number: **Mike or Tom Lucas (209) 277-6499**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Farmer** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
Gioletti	**11888 Fulkerth Road**	**Turlock**	**95380**	**209-678-6603**

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
Stanislaus County			**APN 0022-0037-0003-0000**

Dates Hauled: **3/1/23, 3/9/23, 4/4/23, 7/17/23, 7/30/23 & 8/12/23**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____Tons

Manure Solids Content (if amount reported in tons): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 2,671,500 Gallons

Method used to determine volume of process wastewater: _____

Total Run Time x Pump Output GPM

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mike Lucas (Mar 28, 2024 15:12 PDT) Date: 28/03/24

Hauler's Signature: Mike Lucas (Mar 28, 2024 15:12 PDT) Date: 28/03/24

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

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2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: **Mike or Tom Lucas**

Name of Dairy Facility: **Lucas Dairy**

Facility Address: **11204 Fulkerth Road, Turlock, CA 95380**

Contact Person Name and Phone Number: **Mike or Tom Lucas (209) 277-6499**

Manure Hauler Information:

Name of Hauling Company/Person: **Lucas Dairy**

Address of Hauling Company /Person: **11204 Fulkerth Road, Turlock, CA 95380**

Contact Person and Phone Number: **Mike or Tom Lucas (209) 277-6499**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Farmer** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
Mike Nelson	**694 E. Olive Ave**	**Turlock**	**95380**	**(209) 765-8105**

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
Stanislaus County			**APN 0022-0037-0002-0000**

Dates Hauled: **3/4/23, 3/12/23, 4/7/23, 7/20/23, 8/2/23 & 8/15/23**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____Tons

Manure Solids Content (if amount reported in tons): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **1,560,000** Gallons

Method used to determine volume of process wastewater: _____

Total Run Time x Pump Output GPM

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mike Lucas (Mar 28, 2024 15:12 PDT) Date: 28/03/24

Hauler's Signature: Mike Lucas (Mar 28, 2024 15:12 PDT) Date: 28/03/24

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: **Mike or Tom Lucas**

Name of Dairy Facility: **Lucas Dairy**

Facility Address: **11204 Fulkerth Road, Turlock, CA 95380**

Contact Person Name and Phone Number: **Mike or Tom Lucas (209) 277-6499**

Manure Hauler Information:

Name of Hauling Company/Person: **J & F Fertilizer**

Address of Hauling Company /Person: **900 Meredith Ave. Gustine, CA 95322**

Contact Person and Phone Number: **John Nunes (209) 854-6325**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Broker** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
J & F Fertilizer	900 Meredith Avenue	Gustine	95322	209-854-6325

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
900 Meredith Avenue	Gustine	95322	

Dates Hauled: **10/15/2023**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: **2,700** Tons

Manure Solids Content (if amount reported in tons): **93.7%**

Method used to determine amount of manure: **Number of Truckloads x Average Truck Weight**

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **N/A** Gallons

Method used to determine volume of process wastewater: **N/A**

Written Agreement:

****NOT NECESSARY FOR SOLID MANURE MANIFEST****

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mike Lucas (Mar 28, 2024 15:12 PDT) Date: 28/03/24

Hauler's Signature: John Nunes Date: 10/15/2023

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Lucas Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:14

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0898-01	Blaker	Ag Water	T. Madruga		11/15/2023 12:00
23K0898-02	Dilworth	Ag Water	T. Madruga		11/15/2023 12:30
23K0898-03	Mike	Ag Water	T. Madruga		11/15/2023 13:00
23K0898-04	Milk Barn	Ag Water	T. Madruga		11/15/2023 13:30
23K0898-05	Norman	Ag Water	T. Madruga		11/15/2023 14:00
23K0898-06	Old Milk Barn	Ag Water	T. Madruga		11/15/2023 14:30
23K0898-07	Pit	Ag Water	T. Madruga		11/15/2023 15:00
23K0898-08	Tom	Ag Water	T. Madruga		11/15/2023 15:30

Default Cooler Temperature on Receipt °C: 1.8
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Lucas Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:14

Sample Results

Sample: Blaker
23K0898-01 (Water)

Sampled: 11/15/2023 12:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.16**	mmhos/cm	0.01	1		11/20/23 13:51	SM 2510 B		BEK0616
Electrical Conductivity umhos	**1160**	umhos/cm	10.0	1		11/20/23 13:51	SM 2510 B		BEK0616
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 12:00	Field		BEK0558
Nitrate Nitrogen as NO3N	**43.9**	mg/L	0.1	1	10	11/17/23 01:59	EPA 300.0		BEK0603
Temperature	**25.0**	units	0.0	1		11/20/23 13:51	SM 4500-H+	H	BEK0616
pH	**7.4**	units	1.0	1		11/20/23 13:51	SM 4500-H+	H	BEK0616

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Lucas Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:14

Sample Results
(Continued)

Sample: Dilworth
23K0898-02 (Water)

Sampled: 11/15/2023 12:30
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.33**	mmhos/cm	0.01	1		11/20/23 13:53	SM 2510 B		BEK0616
Electrical Conductivity umhos	**1330**	umhos/cm	10.0	1		11/20/23 13:53	SM 2510 B		BEK0616
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 12:30	Field		BEK0558
Nitrate Nitrogen as NO3N	**29.5**	mg/L	0.1	1	10	11/17/23 02:20	EPA 300.0		BEK0603
Temperature	**25.0**	units	0.0	1		11/20/23 13:53	SM 4500-H+	H	BEK0616
pH	**7.5**	units	1.0	1		11/20/23 13:53	SM 4500-H+	H	BEK0616

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Lucas Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:14

Sample Results
(Continued)

Sample: Mike
23K0898-03 (Water)

Sampled: 11/15/2023 13:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.35**	mmhos/cm	0.01	1		11/20/23 13:54	SM 2510 B		BEK0616
Electrical Conductivity umhos	**1350**	umhos/cm	10.0	1		11/20/23 13:54	SM 2510 B		BEK0616
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 13:00	Field		BEK0558
Nitrate Nitrogen as NO3N	**29.7**	mg/L	0.1	1	10	11/17/23 02:42	EPA 300.0		BEK0603
Temperature	**25.0**	units	0.0	1		11/20/23 13:54	SM 4500-H+	H	BEK0616
pH	**7.4**	units	1.0	1		11/20/23 13:54	SM 4500-H+	H	BEK0616

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Lucas Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:14

Sample Results
(Continued)

Sample: Milk Barn
23K0898-04 (Water)

Sampled: 11/15/2023 13:30
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.33**	mmhos/cm	0.01	1		11/20/23 13:55	SM 2510 B		BEK0616
Electrical Conductivity umhos	**1330**	umhos/cm	10.0	1		11/20/23 13:55	SM 2510 B		BEK0616
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 13:30	Field		BEK0558
Nitrate Nitrogen as NO3N	**41.9**	mg/L	0.1	1	10	11/17/23 03:04	EPA 300.0		BEK0603
Temperature	**25.0**	units	0.0	1		11/20/23 13:55	SM 4500-H+	H	BEK0616
pH	**7.4**	units	1.0	1		11/20/23 13:55	SM 4500-H+	H	BEK0616

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Lucas Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:14

Sample Results
(Continued)

Sample: Norman
23K0898-05 (Water)

Sampled: 11/15/2023 14:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.83**	mmhos/cm	0.01	1		11/20/23 13:57	SM 2510 B		BEK0616
Electrical Conductivity umhos	**834**	umhos/cm	10.0	1		11/20/23 13:57	SM 2510 B		BEK0616
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 14:00	Field		BEK0558
Nitrate Nitrogen as NO3N	**24.8**	mg/L	0.1	1	10	11/17/23 03:26	EPA 300.0		BEK0603
Temperature	**25.0**	units	0.0	1		11/20/23 13:57	SM 4500-H+	H	BEK0616
pH	**7.8**	units	1.0	1		11/20/23 13:57	SM 4500-H+	H	BEK0616

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Lucas Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:14

Sample Results
(Continued)

Sample: Old Milk Barn
23K0898-06 (Water)

Sampled: 11/15/2023 14:30
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.35**	mmhos/cm	0.01	1		11/20/23 13:58	SM 2510 B		BEK0616
Electrical Conductivity umhos	**1350**	umhos/cm	10.0	1		11/20/23 13:58	SM 2510 B		BEK0616
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 14:30	Field		BEK0558
Nitrate Nitrogen as NO3N	**42.0**	mg/L	0.1	1	10	11/17/23 03:47	EPA 300.0		BEK0603
Temperature	**25.0**	units	0.0	1		11/20/23 13:58	SM 4500-H+	H	BEK0616
pH	**7.4**	units	1.0	1		11/20/23 13:58	SM 4500-H+	H	BEK0616

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R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Lucas Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:14

Sample Results
(Continued)

Sample: Pit
23K0898-07 (Water)

Sampled: 11/15/2023 15:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**2.02**	mmhos/cm	0.01	1		11/20/23 13:59	SM 2510 B		BEK0616
Electrical Conductivity umhos	**2020**	umhos/cm	10.0	1		11/20/23 13:59	SM 2510 B		BEK0616
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 15:00	Field		BEK0558
Nitrate Nitrogen as NO3N	**89.9**	mg/L	0.1	1	10	11/17/23 04:09	EPA 300.0		BEK0603
Temperature	**25.0**	units	0.0	1		11/20/23 13:59	SM 4500-H+	H	BEK0616
pH	**7.6**	units	1.0	1		11/20/23 13:59	SM 4500-H+	H	BEK0616

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Lucas Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:14

Sample Results
(Continued)

Sample: Tom
23K0898-08 (Water)

Sampled: 11/15/2023 15:30
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.16**	mmhos/cm	0.01	1		11/20/23 14:01	SM 2510 B		BEK0616
Electrical Conductivity umhos	**1160**	umhos/cm	10.0	1		11/20/23 14:01	SM 2510 B		BEK0616
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 15:30	Field		BEK0558
Nitrate Nitrogen as NO3N	**18.0**	mg/L	0.1	1	10	11/17/23 04:31	EPA 300.0		BEK0603
Temperature	**25.0**	units	0.0	1		11/20/23 14:01	SM 4500-H+	H	BEK0616
pH	**7.5**	units	1.0	1		11/20/23 14:01	SM 4500-H+	H	BEK0616

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LABORATORY INC

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8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Lucas Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:14

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0603										
Blank (BEK0603-BLK1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0603-BLK2)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0603-BLK3)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0603-BS1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.2	90-110		
LCS (BEK0603-BS2)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	4.8		0.1	mg/L	5.000		96.8	90-110		
Duplicate (BEK0603-DUP1)		**Source: 23K0892-03**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	10.8		0.1	mg/L		10.8			0.157	10
Duplicate (BEK0603-DUP2)		**Source: 23K0898-05**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	25.4		0.1	mg/L		24.8			2.19	10
Matrix Spike (BEK0603-MS1)		**Source: 23K0892-03**			Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	15.9		0.1	mg/L	5.000	10.8	101	90-110		
Matrix Spike (BEK0603-MS2)		**Source: 23K0898-05**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	30.4		0.1	mg/L	5.000	24.8	112	90-110		
Reference (BEK0603-SRM1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	9.8			mg/L	10.00		97.9	90-110		
Reference (BEK0603-SRM2)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	9.7			mg/L	10.00		97.1	90-110		
Reference (BEK0603-SRM3)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	9.8			mg/L	10.00		97.9	90-110		

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Lucas Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:14

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0616										
Blank (BEK0616-BLK1)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.7		1.0	units						
Blank (BEK0616-BLK2)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.2		1.0	units						
Blank (BEK0616-BLK3)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.7		1.0	units						
Duplicate (BEK0616-DUP1)		**Source: 23K0896-03**			Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	0.47		0.01	mmhos/cm		0.47			0.423	10
pH	7.8		1.0	units		7.8			0.255	10
Electrical Conductivity umhos	472		10.0	umhos/cm		474			0.423	10
Duplicate (BEK0616-DUP2)		**Source: 23K0899-01**			Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1.65		0.01	mmhos/cm		1.64			0.298	10
Electrical Conductivity umhos	1650		10.0	umhos/cm		1640			0.298	10
pH	7.0		1.0	units		7.0			0.00	10
Reference (BEK0616-SRM1)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	561			umhos/cm	538.0		104	90-110		
Reference (BEK0616-SRM2)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	7.6			units	7.520		101	67021-101.32		
Reference (BEK0616-SRM3)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEK0616-SRM4)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEK0616-SRM5)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		

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LABORATORY INC

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Hilmar, CA 95324

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Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Lucas Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:14

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0616 (Continued)										
Reference (BEK0616-SRM5)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEK0616-SRM6)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEK0616-SRM7)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEK0616-SRM8)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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11/16/23 15:35 23K0898

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 8 No. of Bottles: 8

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Results Need By

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State: CA** **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:

REQUESTED BY: Rui & John Avila
PROJECT Lucas Dairy
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	Blaker	11/15/2023	1200	1.8	ND
2	Dilworth	11/15/2023	1230	2.3	ND
3	Mike	11/15/2023	1300	1.6	ND
4	Milk Barn	11/15/2023	1330	.9	ND
5	Norman	11/15/2023	1400	1.6	ND
6	Old Milk Barn	11/15/2023	1430	2.1	ND
7	Pit	11/15/2023	1500	2.6	ND
8	Tom	11/15/2023	1530	1.8	ND

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/15/2023 @ 1530	11/16/2023 @9am
Second	JLF	Dellavalle Laboratory, Inc.	11/16/2023 @9am	11-16-23 3:25
Third		DLI	11.16.23 15:35	
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/16/23 15:35 23K0898

Shipping Information: Fed Ex □ UPS □ GSO □ Picked-Up ☑ Walk In □ DLI Sampler □
Ontrac □ PMS □ USPS □ Other □ _____

Shipping Container: Ice Chest ☑ None □ Box □

Refrigerant: Wet Ice: ☑ Blue Ice: □ None: □

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1	1	1	1	1	1	1		
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										